

Online Examination Report

Date:	14.08.2014
Attempts:	1
Examinee:	Abey, Stacy
Date of birth:	10/10/1988
Subject:	080- Principles of flight
Result:	34,99 %
Time used:	00:21:20 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	188	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	205	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
3	222	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
4	220	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
5	209	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
6	199	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
7	198	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	219	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	146	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
10	149	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
11	204	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
12	186	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
13	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
14	212	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
15	208	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
16	216	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
17	25	81.1.1.1	Laws and definitions	0,00	1,00
18	30	81.1.2	Two-dimensional airflow around an aerofoil	0,00	1,00
19	95	81.1.2.2	Stagnation point	0,00	1,00
20	27	81.1.2.2	Stagnation point	0,00	1,00
21	34	81.1.4.2	Induced drag	0,00	1,00
22	56	81.1.6	Ground effect	0,00	1,00
23	44	81.1.8.5	Special phenomena of stall	0,00	1,00
24	28	81.1.8.5	Special phenomena of stall	1,00	1,00
25	78	81.1.12.1	Ice and other contaminants	1,00	1,00
26	36	81.2.2	Shock waves	0,00	1,00
27	16	81.2.2.2	Oblique shock waves	0,00	1,00
28	45	81.2.4	Buffet onset	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	102	81.4	STABILITY	0,00	1,00
30	10	81.5.8.3	Stabiliser trim	0,00	1,00
31	32	81.5.8.3	Stabiliser trim	0,00	1,00
32	6	81.6.1.1	Flutter	1,00	1,00
33	96	81.6.1.1	Flutter	0,00	1,00
34	94	81.6.2.1	Manoeuvring load diagram	0,00	1,00
35	112	81.7	PROPELLERS	1,00	1,00
36	106	81.7	PROPELLERS	0,00	1,00
37	76	81.7.1.1	Relevant propeller parameters	1,00	1,00
38	51	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
39	8	81.7.4.2	Gyroscopic precession	0,00	1,00
40	29	81.8.1.4	Straight steady glide	0,00	1,00
Total:35%				14,00	40,00

Attachment 2: List of result by topic

Licence

Date: 14.08.2014

Examinee: Abey, Stacy

Location: Uganda Civil Aviation Authority

Your Result: 14,00 from 40,00 Points (35,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	14,00	40,00	35,00
81.1. SUBSONIC AERODYNAMICS	9	2,00	9,00	22,22
81.2. HIGH SPEED AERODYNAMICS	3	0,00	3,00	0,00
81.4. STABILITY	1	0,00	1,00	0,00
81.5. CONTROL	2	0,00	2,00	0,00
81.6. LIMITATIONS	3	1,00	3,00	33,33
81.7. PROPELLERS	5	2,00	5,00	40,00
81.8. FLIGHT MECHANICS	1	0,00	1,00	0,00
Total	40	14,00	40,00	35,00

Candidate: Stacy Abey
Date of birth: 1988-10-10

Subject: 080- Principles of flight
Date: 2014-08-14



1

In a twin-engine airplane, the single-engine service ceiling is the maximum density altitude at which V(YSE) will produce

☐ 100 feet per minute rate of climb.

☒ 500 feet per minute rate of climb.

☐ 50 feet per minute rate of climb.

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2

What could result if a glider pilot releases while in the low-tow position during an aerotow?

- ☐ Nose of the glider would tend to pitch up after release.
- ☐ Glider may be forced into the towplane's wake turbulence.
- ☒ Tow ring may strike and damage the glider after release.

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3

Which stall must be performed during a flight instructor - airplane practical test?

☒ Power-on or power-off.

☐ Imminent.

☐ Accelerated.

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4

What is one procedure for relighting the burner while in flight?

- ☐ Open the blast valve full open and light the pilot light.
- ☐ Close the tank valves, vent the fuel lines, reopen the tank valves, and light the pilot light.
- ☒ Open another tank valve, open the blast valve, and light the main jet using reduced flow.

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5

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- ☐ The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.
- ☐ The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.
- ☒ The desired landing spot would be undershot or the glider would stall.

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The best lift/drag ratio of a glider is a value that

- ☐ remains constant regardless of airspeed changes.
- ☒ remains constant and is independent of the weight being carried.
- ☐ varies depending upon the weight being carried.

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A pilot of a high-performance airplane should be aware that flying a steeper-than-normal VASI glide slope angle may result in

☐ increased landing rollout.

☒ a hard landing.

☐ landing short of the runway threshold.

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How should a roundout from a moderate-rate ascent to level flight be made?

- ☐ Reduce the amount of heat gradually as the balloon approaches altitude.
- ☒ Cool the envelope by venting and add heat just before arriving at the desired altitude.
- ☐ Vent at altitude and add heat upon settling back down to altitude.

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Which statement is generally true regarding wing camber of a glider's airfoil?

- ☐ There is no camber on either the upper or lower surface of the wing.
- ☒ The camber is greater on the upper wing surface than it is on the lower surface of the wing.
- ☐ The camber is the same on both the upper and lower surface of the wing.

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At what bank angle will the resultant of gravity and centrifugal force equal twice a glider's weight?

☒ 30 degrees.

☐ 60 degree.

☐ 45 degrees.

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How can excessive towline slack that is allowed to develop during a glider tow be eliminated?

- ☐ Execute a shallow banked coordinated turn to either side.
- ☒ Yaw the nose to one side with rudder while keeping the wings level with ailerons.
- ☐ Increase pitch attitude until towline becomes taut.

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The center of gravity of an aircraft is computed along the

☒ lateral axis.

☐ longitudinal axis.

☐ vertical axis.

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13

What is the suggested speed to fly when passing through lift with no intention to work the lift?

☒ Minimum sink speed.

☐ Best lift/drag speed.

☐ Best glide speed.

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14

With regard to two or more gliders flying in the same thermal, which statement is true?

- ☒ All turns should be to the right.
- ☐ Turns should be made in the same direction as the first glider to enter the thermal.
- ☐ Turns should be in the same direction as the highest glider.

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15

During a winch launch, which factor would most likely result in pitch oscillations?

- ☐ Winching speed too slow.
- ☐ Winching speed too fast.
- ☒ Insufficient up-elevator control.

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It may be possible to make changes in the direction of flight of a hot air balloon by

☒ flying a constant atmospheric pressure gradient.

☐ operating at different flight altitudes.

☐ using the maneuvering vent.

17

Which of these statements about a stationary subsonic flow are correct or incorrect?

- I. The static pressure decreases as the streamlines converge.
II. The velocity increases as the streamlines converge.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☒ I is correct, II is incorrect.

☐ I is correct, II is correct.

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(Please view annex 081-0005 issue date July 2004)

How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

☒ $V_1 > V_2$ and $V_2 < V$.

☐ $V_1 = 0$ and $V_2 > V$.

☐ $V_1 = 0$ and $V_2 = V$.

☐ $V_1 < V_2$ and $V_2 < V$.

See Attachments:

1. 081-0005.gif

Annex

(Attachment
No.1)

19

Assuming no flow separation, which of these statements about the flow around an aerofoil as the angle of attack increases are correct or incorrect?

I. The stagnation point moves down.

II. The point of lowest static pressure moves aft.

☐ I is correct, II is incorrect.

☐ I is correct, II is correct.

☒ I is incorrect, II is correct.

☐ I is incorrect, II is incorrect.

20

The stagnation point is the point:

- ☐ of the intersection of the total aerodynamic force and the chord line.
- ☒ relative to which the sum of all moments is independent of angle of attack.
- ☐ where the velocity of the relative airflow is reduced to zero.
- ☐ of the intersection of the thrust vector and the chord line.

21

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates less induced drag than a rectangular lift distribution.
- II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

22

Which statement, about an aeroplane entering ground effect at constant angle of attack, is correct?

I. The lift coefficient C_L increases.

II. The induced drag coefficient C_{Di} increases.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☒ I is correct, II is incorrect.

☒ I is correct, II is correct.

23

Regarding deep stall characteristics, identify whether the following statements are correct or incorrect:

I. The combination of a wing with sweepback and a T-tail make an aeroplane prone to deep stall.

II. A stick pusher system can be fitted to an aeroplane that exhibits abnormal stall characteristics.

☐ I is correct, II is correct.

☒ I is correct, II is incorrect.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

24

A stick pusher:

- ☒ pushes the elevator control forward when a specified value of angle of attack is exceeded.
- ☐ vibrates the elevator control.
- ☐ pushes the elevator control to avoid a stall at a negative load factor.
- ☐ pushes the elevator control forward prior to stick shaker activation.

25

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

- ☒ **decrease of CLMAX and increase of drag.**
- ☐ **increase of CLMAX and increase of drag.**
- ☐ **increase of CLMAX and decrease of drag.**
- ☐ **decrease of CLMAX and decrease of drag.**

26

Which of these statements about critical mach.No. is correct?

- ☐ Flight at any speed above critical mach no. causes severe vibration of the aeroplane.
- ☐ critical mach. No. is always greater than 1.
- ☒ Shock waves cannot occur at speeds below critical mach no.
- ☒ As speed increases above critical mach No., parasite drag decreases rapidly.

27

Compared with an oblique shock wave a normal shock wave has a:

- ☒ higher total pressure.
- ☐ lower static temperature.
- ☐ higher loss in total pressure.
- ☐ higher total temperature.

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28

What is the significance of the maximum allowed cruising altitude, based on the 1.3 g margin?

- ☒ a manoeuvre with a load factor of 1.3 will cause Mcrit to be exceeded.
- ☐ exceeding a load factor of 1.3 will cause permanent deformation of this aeroplane.
- ☐ a manoeuvre with a load factor of 1.3 will cause a Mach number at which accelerated low speed stall occurs.
- ☐ a manoeuvre with a load factor of 1.3 will cause buffet onset.

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29

The capability of an aircraft to respond to a pilot's inputs, especially with regard to flightpath and attitude, is

☐ controllability.

☒ response.

☐ maneuverability.

30

Comparing the differences between a horizontal trimmable stabiliser and an elevator trim tab, which statement is correct?

- I. A stabiliser trim is less suitable for jet transport aeroplanes because of their large speed range.
II. A trim tab is a more powerful means of trimming.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☒ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

31

When comparing a stabiliser trim system with an elevator trim system, which of these statements is correct?

- ☒ **an elevator trim is more suitable for aeroplanes with a large CG range.**
- ☐ **a stabiliser trim is more sensitive to flutter.**
- ☐ **a stabiliser trim is able to compensate larger changes in pitching moments.**
- ☐ **an elevator trim is able to compensate larger changes in pitching moments.**

32

Which of these statements about flutter are correct or incorrect?

I. If flutter occurs, IAS should be reduced.

II. Resistance to flutter increases with increasing wing stiffness.

☒ I is correct, II is correct.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

33

Wing flutter may be caused by a:

- ☐ combination of fuselage bending and wing torsion.
- ☒ aerodynamic wing stall at high speed.
- ☐ combination of bending and torsion of the wing structure.
- ☐ combination roll control reversal and low speed stall.

34

The relationship between the stall speed V_S and V_A (EAS) for a large transport aeroplane can be expressed in the following formula:

☐ $V_A > V_S * \text{SQRT}(2.5).$

☒ $V_A < V_S * \text{SQRT}(2.5).$

☐ $V_A \leq V_S * \text{SQRT}(2.5).$

☐ $V_A \geq V_S * \text{SQRT}(2.5).$

35

On a multiengine airplane, where the propellers rotate in the same direction, why is the loss of power on one engine more critical than the loss of power on the other engine?

- ☐ The torque reaction from operation of the critical engine is more severe around the vertical axis as well as the longitudinal axis.
- ☐ The corkscrew pattern of airflow from one propeller is less effective against the airflow from the critical engine.
- ☒ The asymmetric propeller thrust or P-factor results in the center of thrust from one engine being farther from the airplane centerline than the center of thrust from the other engine.

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Propeller slip is the difference between the

- ☐ geometric pitch and blade angle of the propeller.
- ☒ plane of rotation of the propeller and forward velocity of the aircraft.
- ☐ geometric pitch and the effective pitch of the propeller.

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The difference between a propeller's blade angle and its angle of attack is called:

- ☐ the effective pitch.
- ☐ the propeller angle.
- ☒ the helix angle.
- ☐ propeller slip.

38

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A fixed pitch propeller improves propeller efficiency over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☒ I is correct, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.

39

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces left yaw.
II. Right yaw produces pitch up.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☒ I is correct, II is correct.

☐ I is correct, II is incorrect.

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40

Which of the following increases the maximum duration of a glide?

☐ An increase in mass.

☒ A tailwind.

☐ A headwind.

☐ A decrease in mass.

Attachment No. 1

Questions 18

Annex

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081-0005

